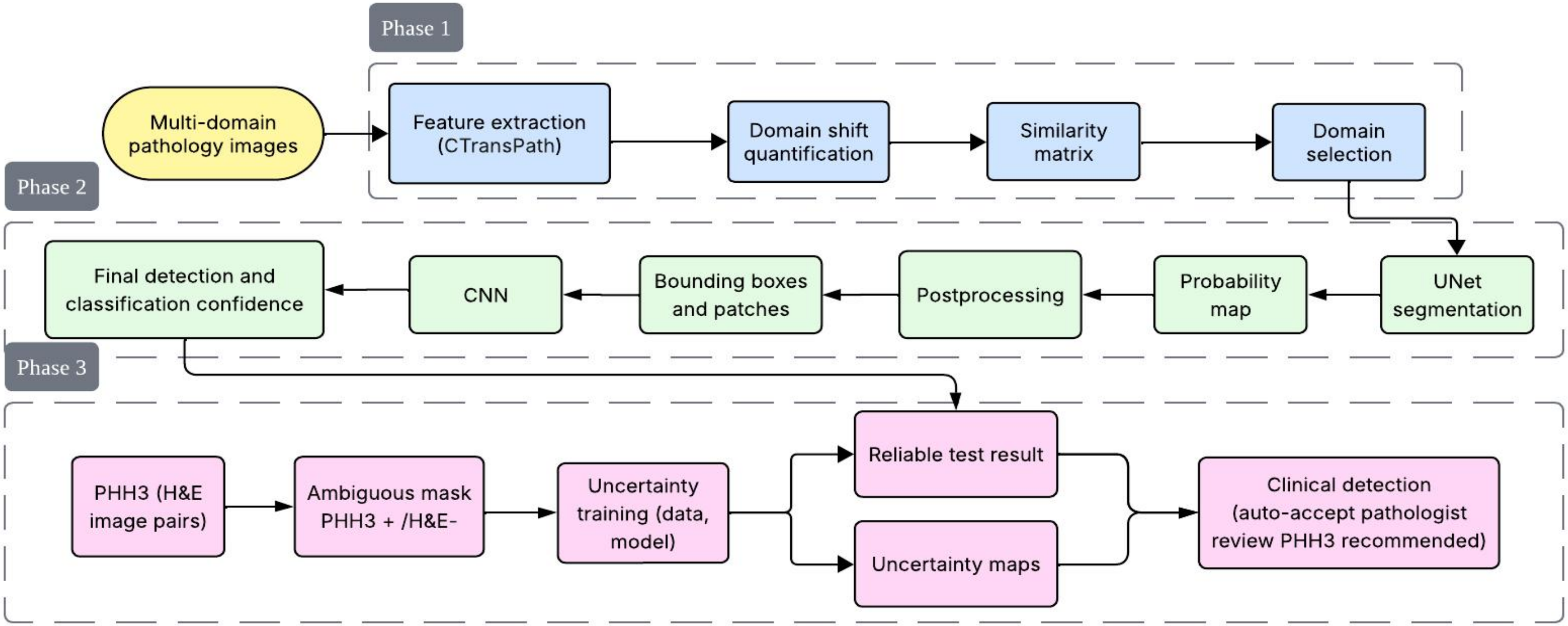
Presenter: Chaohui Xu

Members: Yang Cao, Naina Misra, Lillian Lam

Recap pipeline



Immunohistochemical Marker (IHC)

Ki67, PHH3 for ambiguous modeling

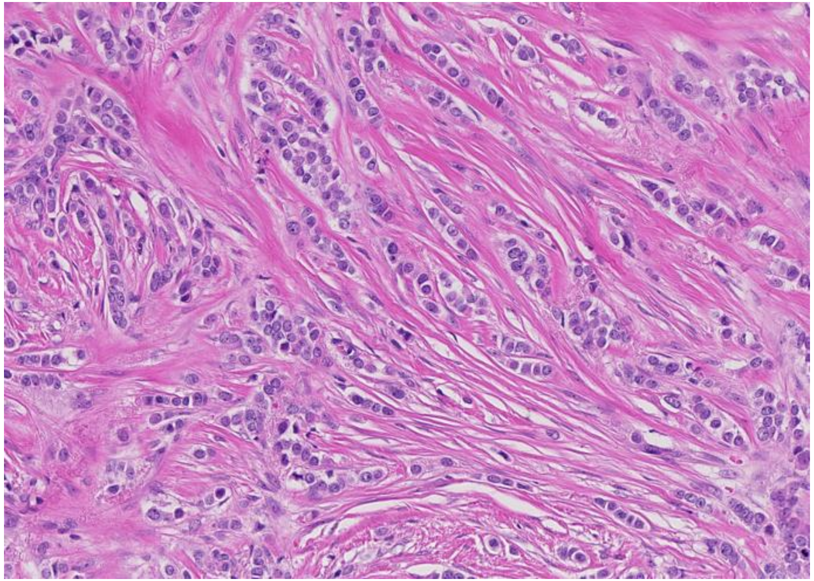
Why focus on proliferation indicators?

Mitotic count matters:

- Essential prognostic indicator
- Higher mitotic activity → more aggressive tumor behavior
- Used in tumor grading systems

Limitation of H&E for mitosis

- High subjectivity
- Morphological Variability
- Overlapping nuclei and artifacts



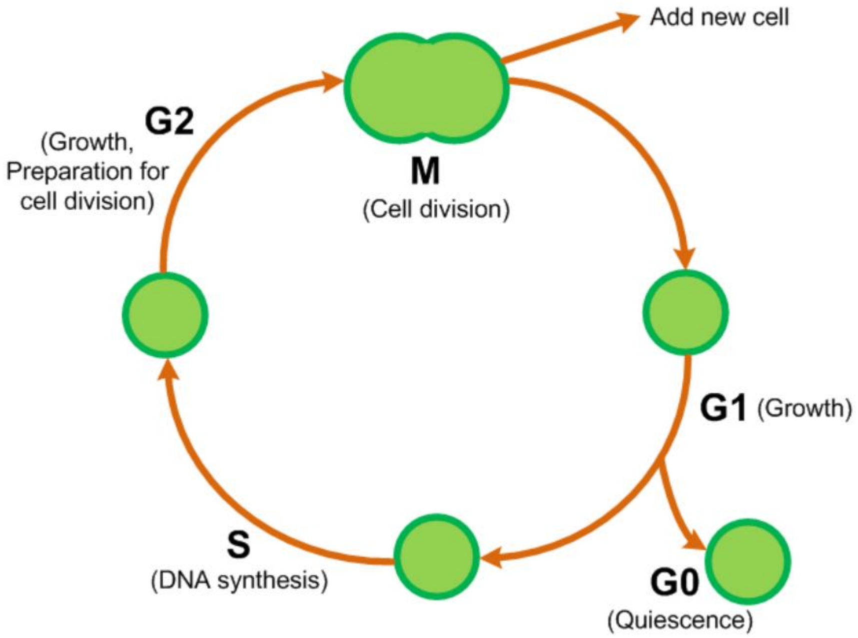
Ki67

Rule of Ki67

- Marker of cell proliferation
- Expressed in G1, S, G2 and M phase

Limitations

- Not mitosis-specific
- Stain intensity varies across labs



Ki67

Appearance

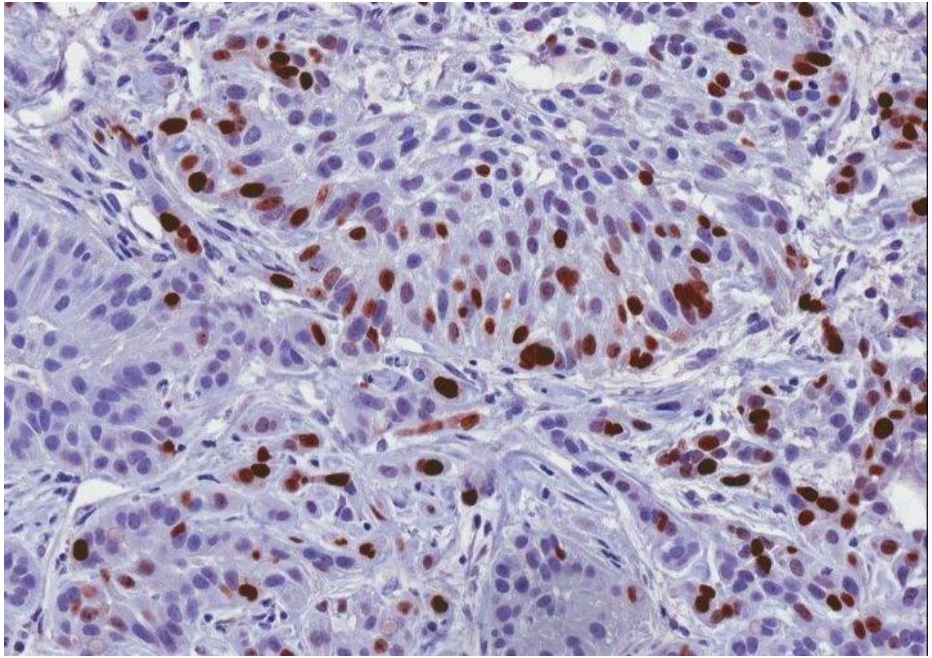
- Positive cell: brown nuclear staining
- Highlights spatial heterogeneity

Relevance for MIDOG++

- Distinguish active tumor areas
- Auxiliary label or weak supervision

Lighter stain - G1 and S

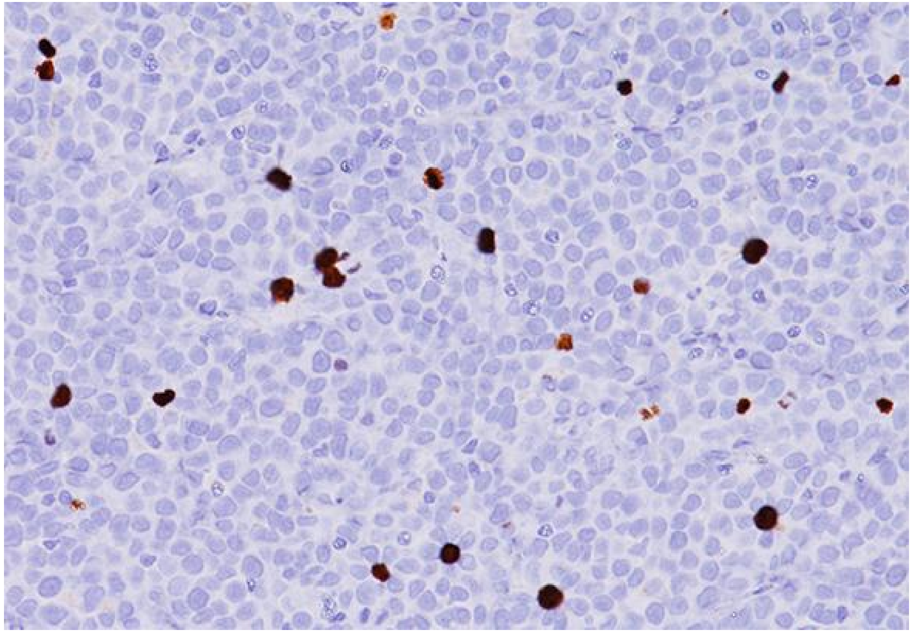
Deeper stain - prophase (starting mitosis)



PHH3 (Phospho-Histone H3)

Rule of PHH3

- Marker of mitosis
- Expressed only in late G2 and M phase



Advantages

- Mitosis specific
- Very low false positive
- High quality reference stain

PHH3 (Phospho-Histone H3)

Relevance for MIDOG++

- Reliable mitosis annotations for training
- Helps evaluate model robustness across stains
- Useful for multi-stain learning (H&E → PHH3 transfer)
- Better consistency across domains

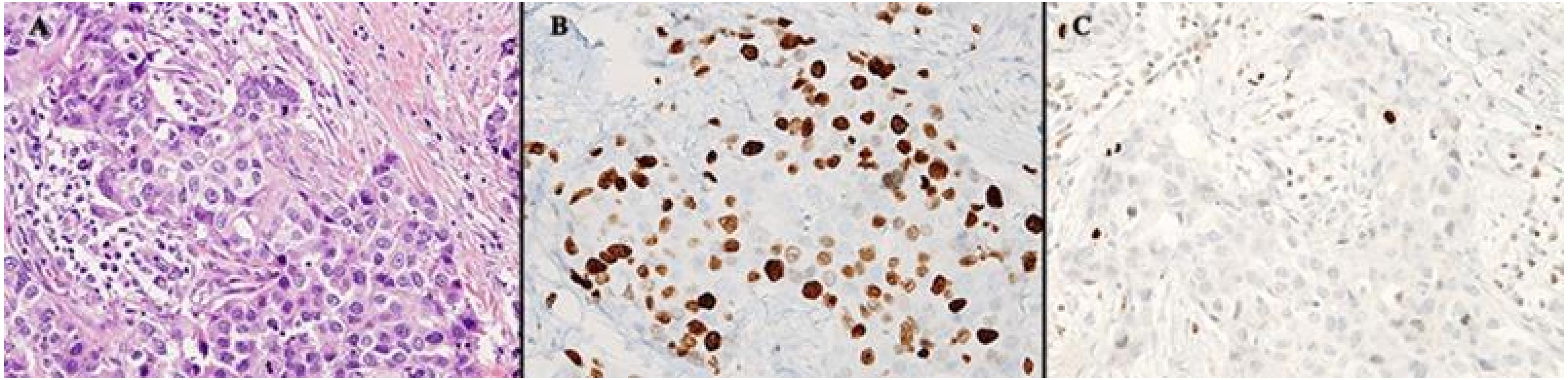
Challenges

- Different labs may produce different PHH3 intensities
- Requires domain adaptation or stain normalization

Ki67 vs PHH3

Ki67	PHH3
Proliferation marker	Mitosis marker
G1/S/G2/M	Late G2 → M only
Auxiliary signal	Main ground-truth label

H&E Ki67 (IHC) PHH3 (IHC)



Limitations

There are many datasets and some open source for Ki67

Dataset	Source	Image Type	Ground Truth	Quantity	Scanner	Mag	Tissue
SMH Patches	St. Michael's Hospital	Patches	Individual Nuclei	Ideal: 330 Non Ideal: 330	Aperio AT Turbo	x20	Human Breast
SMH WSIs	St. Michael's Hospital	WSI	Proliferation Index	55	Aperio AT Turbo	x20	Human Breast
DeepSlides	Senaras et al. (Open Source)	Patches	Individual Nuclei	452	Aperio ScanScope	x40	Human Breast
Protein Atlas	Protein Atlas (Open Source)	TMA	Proliferation Index Ranges	56	Aperio AT Turbo & Aperio T2	x20	Human Breast
OVC	Ontario Veterinary College	TMA	Individual Nuclei	30	Leica SCN400	x20	Canine Mammary

PHH3 is more rare and private

Other immunohistochemical (IHC) Markers?

- MPM-2: stains the cytoplasm for early prophase, metaphase, anaphase and telophase
- pSlugS158: More specificity for mitosis than PHH3 (less G2 staining)

Issue: No Large Datasets!

Next Step

- Reaching out for Ki67 and PHH3 ground-truth dataset
- Classifying only metaphase, anaphase, and telophase in Ki67
- Domain shift similarity with CORAL
- Start training with subset of WSIs

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Thank you